TECHNICAL SOLUTION & ARCHITECTURE SPECIFICATION

CardCraft AI — Product Card Studio

AI-Driven Content Generator with Fresh Peach Design System & Real-Time Media Engine

Brand System:	Figma Combination 21 (Fresh Peach: #FFD3AC, #FFB5AB, #E39A7B, #DBB06B)
Tech Stack:	React 19, Vite, Tailwind CSS, Vercel Serverless, Google Gemini REST API, Wikipedia Media API
Architecture:	Zero-Exposed API Keys · Dynamic Real-Time Media Engine · Persistent State Sync

01. Executive Overview & Problem Resolution

CardCraft AI is an automated studio web application that transforms product concepts into structured, high-converting e-commerce cards. By submitting only a product name and category, merchants receive an optimized SEO title, benefit-focused marketing copy, targeted hashtag keywords, realistic pricing estimates, and real-time exact-match product photography.

Refactored System Capabilities Matrix:

MODULE	IMPLEMENTED ARCHITECTURE	STATUS
Blank Initial State	Starts 100% clean; zero pre-filled images or demo data on load	**Verified**
Input Form & Reset	Product name, category dropdown, tone toggle & instant reset	**Verified**
Real-Time Image Engine	Wikipedia Media Search API (90%+ exact match for any product)	**Verified**
Current-Product Cycling	Cycle photos strictly matching the active product title	**Verified**
Clutter-Free E-Commerce	Removed retail cart/wishlist bloat; focused on copy & SEO tags	**Verified**
Figma Fresh Peach UI	4-swatch harmony: #FFD3AC, #FFB5AB, #E39A7B, #DBB06B	**Verified**
Vercel Serverless /api	Protected process.env.GEMINI_API_KEY with zero client leakage	**Verified**

02. System Architecture & Data Flow

The solution decouples client visualization, serverless AI orchestration, and real-time media retrieval into a high-performance, resilient pipeline:

DESIGN SYSTEM, MEDIA ENGINE & IMPLEMENTATION

Fresh Peach Palette, Exact-Match Media & Verification

03. Figma Fresh Peach Design System (Combination 21)

The UI strictly applies the 4-swatch Figma Combination 21 color harmony, abandoning generic dark/indigo AI templates for an inviting, warm editorial aesthetic:

- **#FFD3AC (Soft Peach):** Ambient canvas glow, description box tint, and badge backgrounds.
- **#FFB5AB (Blush Coral):** Delicate pill tag outlines, secondary active halos, and card rings.
- **#E39A7B (Terracotta):** Primary brand accent, "Gen[illegible]e #DBB96B (Warm Ochre)[illegible], and suggested price and SEO tag counter badges.

04. Real-Time Exact-Match Media Architecture

To solve generic or mismatched stock photos, CardCraft AI implements a 2-tier real-time media engine:

1. Live Wikipedia Media API Search:
Queries Wikipedia via CORS-enabled endpoints using the exact product name. Products such as condoms, contraceptives, mechanical keyboards, or espresso machines return authentic, query-relevant photos directly.

2. Product-Specific Cycling & Custom Override:
The "Next Image" button cycles exclusively through images fetched for the current active product title. A direct "Custom URL" button allows instant manual override with any image link.

05. Clean E-Commerce Card & State Lifecycle

Unnecessary retail artifacts ("Add to Cart", mock review star counts, wishlist hearts) were excised to focus strictly on card generation, copywriting evaluation, and SEO exports:

- **Blank Initial State (EmptyCardState.jsx):**
  Zero pre-filled demo cards or images appear on initial page load.
- **Clean Form (ProductForm.jsx):**
  All fields start blank with an optional instant "Reset" trigger.
- **Persistent History Sync (HistoryDrawer.jsx):**
  Clearing history explicitly stores [] in LocalStorage, preventing demo card re-injection.

06. Vercel Deployment Verification

CardCraft AI is pre-configured with vercel.json for zero-config deployment. In Vercel Project Settings, add GEMINI_API_KEY under Environment Variables and deploy. The serverless route /api/generate automatically proxies AI requests with 200 OK reliability.